

Math 242, S15
Quiz 1

Name: Answer
SSN: xxx-xx-

Instructions: There are totally 2 problems. You must show all of your work to receive a credit for a correct answer.

Problem 1. (5 points) Given a first-order differential equation

$$y' = e^{-2y}.$$

Verify that $y(x) = \frac{1}{2} \ln(2x + C)$ defines a one-parameter family of solutions of the given differential equation, then determine a value of the constant C so that $y(x)$ satisfies the initial condition $y(1) = 0$.

Solution: $y(x) = \frac{1}{2} \ln(2x + C)$

$$\Rightarrow y' = \frac{1}{2} \cdot \frac{1}{2x+C} \cdot 2 = \frac{1}{2x+C}$$
$$e^{-2y} = e^{-2 \cdot \frac{1}{2} \ln(2x+C)} = e^{-\ln(2x+C)} = (2x+C)^{-1} = \frac{1}{2x+C}$$
$$\Rightarrow y' = e^{-2y}$$
$$y(1) = 0 \Rightarrow \frac{1}{2} \ln(2 \cdot 1 + C) = 0$$
$$\Rightarrow \ln(2+C) = 0$$
$$\Rightarrow 2+C = 1 \Rightarrow C = -1$$
$$\Rightarrow y(x) = \frac{1}{2} \ln(2x-1)$$

Problem 2. (5 points) Find a general solution for the first-order differential equations

$$\frac{dy}{dx} = \frac{1}{\sqrt{3x+2}}.$$

$$y(x) = \int \frac{1}{\sqrt{3x+2}} dx$$
$$= \int (3x+2)^{-\frac{1}{2}} dx$$
$$= \frac{2}{3} (3x+2)^{\frac{1}{2}} + C$$